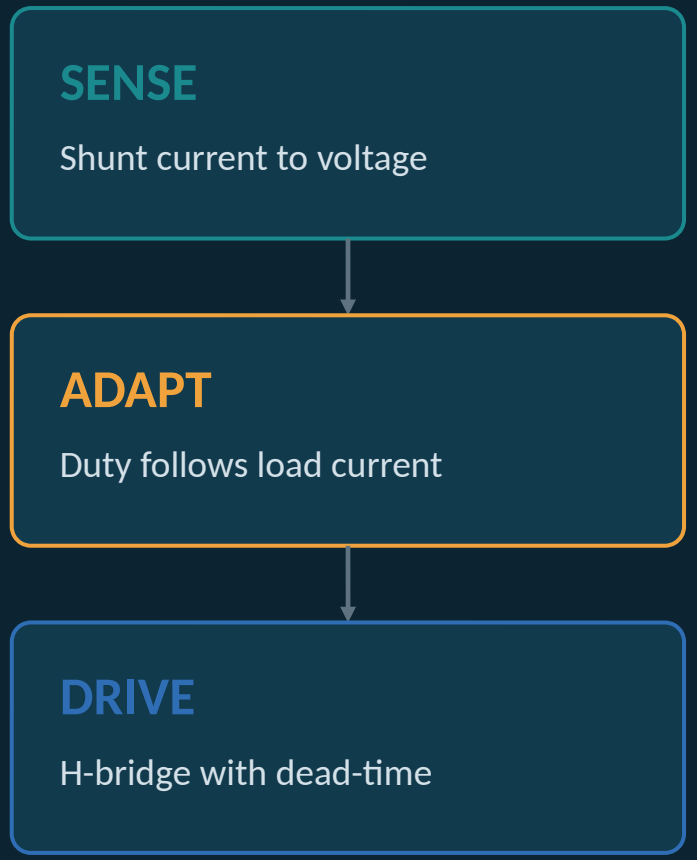
Load-Aware Adaptive H-Bridge Motor Driver for Robotic Actuators

Final Project Review | 23 October 2026

Ashmit Singh
B.Tech Electronics Engineering (VLSI Design & Technology), VIT Vellore
Faculty: Anil Kumar Rajput



Why robotic actuators need an adaptive driver

0.6-1.7 A

motor current range in the adaptive run

1.8 V

gpdk090 simulation supply

85→35 %

PWM duty as load rises (read from plot)

8

instances under the top-level cell

The problem

- Robot joints see variable, unpredictable loads
- Fixed-duty PWM ignores the load current
- Current limits are fixed or used only for protection
- Adaptive ideas (e.g. dead-time) are studied separately from load-aware control

Our idea

- Sense motor current with a shunt and amplifier
- Lower the duty as load current rises, before the hard overcurrent trip
- Keep dead-time and UVLO / OCP protection in one hierarchy

How motor-driver architectures evolved

1 Discrete H-bridge

Conventional

Strength

Flexible, high-current, easy to prototype

Limit

Many external parts; no adaptive features

Typical use

Prototyping, high power

2 Integrated IC

Monolithic

Strength

Compact, protection built in

Limit

Fixed behavior; basic or no current sensing

Typical use

Consumer, general purpose

3 Current-regulated

Closed loop

Strength

Limits current, better torque control

Limit

Fixed current limit; slow to adapt

Typical use

Precision torque, stepper, BLDC

4 Smart / adaptive

Proposed

Strength

Adaptive PWM from load current; integrated sensing and protection

Limit

More complex design and verification

Typical use

Robotics, variable-load actuators

Five papers reviewed

#	Paper	Technology	Supply	Current	Main contribution
1	**Zhang et al., 2010** Fully integrated H-bridge DC motor driver	0.5 um HV CMOS	2.5-5.5 V	1 A	Bridge, control and UVLO / OCP / OTP on one chip
2	**Xu et al., 2012** Monolithic BLDC vibration-motor driver with Hall sensor	0.18 um CMOS	2-4 V	300 mA	Sensor, control and driver on one chip
3	**Krenzke et al., 2003** 36 V H-bridge driver interface	0.35 um CMOS	up to 36 V	external FETs	Charge-pump high-side drive, dead-time, short-circuit detection
4	**Dong et al., 2025** Stepper driver with integrated current sensing	0.18 um BCD	8-40 V	2 A (1.6 A cont.)	Cascode-mirror sensing for PWM current chopping
5	**He et al., 2026** Adaptive dead-time high-side gate driver	0.18 um BCD	24 V input	gate driver	Adaptive dead-time; 7.4 % lower power than fixed dead-time

Specifications as summarized from the papers in the review slides.

Papers 1-3: integration, sensing and high-side drive

Paper 1 | Zhang, 2010

Fully integrated H-bridge

2.5-5.5 V, 1 A, PWM and analog speed control

What we take

Integrating power stage, control and UVLO / OCP / OTP in one IC is practical; gives baseline metrics.

Gap

Conventional control; basic current sensing; no adaptation.

Paper 2 | Xu, 2012

Monolithic BLDC driver with Hall sensor

2-4 V, 300 mA, 7-stage inverter gate drive

What we take

On-chip sensing can sit beside the driver; strong multi-stage gate drive.

Gap

300 mA only; BLDC-specific; no load-dependent drive.

Paper 3 | Krenzke, 2003

36 V H-bridge driver interface

up to 36 V, 50 kHz PWM, on-chip charge pump

What we take

Charge pump, dead-time and cross-conduction protection inform our GATE_DRIVER.

Gap

Gate driver only; external FETs; sensing only for protection.

Papers 4-5: current sensing and adaptive dead-time

Paper 4 | Dong, 2025

8-40 V | 2 A

Integrated current sensing in a stepper driver

Technique

Cascode current mirrors sample the bridge current for PWM current chopping, without an external sense resistor.

What we take

Supports our CURRENT_SENSE block; shows on-chip sensing enables closed-loop control.

Gap

Stepper-specific; regulation is not load-adaptive.

Paper 5 | He, 2026

-7.4 % power

Adaptive dead-time high-side gate driver

Technique

Adaptive dead-time with cross-interlocked drive; verified by tape-out and hardware test (0.18 um BCD, 24 V).

What we take

Techniques reusable in our gate driver; shows adaptation pays off.

Gap

No load-current-based duty control.

Base paper: Dong et al., IEICE Electronics Express, 2025

What the base paper gives

- Two-phase stepper driver in 0.18 um BCD
- 8-40 V supply; 2 A maximum, 1.6 A continuous
- Integrated H-bridge, driver circuits and protection
- Cascode current mirrors sample the bridge current

What was missing for us

- Stepper only: no brushed-DC actuator case
- Regulation is reference-based current chopping, not load-aware PWM
- No explicit gate-driver and dead-time hierarchy
- Sensing serves chopping; duty is not reduced as load rises

What we built on top

- Brushed-DC robotic actuator in one Cadence hierarchy
- Shunt and sense amplifier feeding ISENSE_OUT
- Current-feedback adaptive PWM: duty falls as load current rises
- Gate_Driver with a Dead_Time cell per leg; UVLO and OCP in one FAULT path

Supporting reference Paper 1 (2010), fully integrated H-bridge DC-motor driver: PWM / analog speed control and UVLO, OCP, OTP protection. It backs our DC-motor power stage and protection architecture. Our work extends these papers; it is not a claim of absolute novelty.

Base paper versus our extension

Feature	Dong et al., 2025 (base paper)	Our extension
Motor application	Two-phase stepper motor	Brushed-DC robotic actuator
Power stage	Integrated H-bridge, 0.18 um BCD, 8-40 V, 2 A	H-bridge with explicit Gate_Driver and a Dead_Time cell per leg (gpdk090, 1.8 V simulated)
Current sensing	Integrated cascode current mirrors	Shunt resistor and sense amplifier
Current control	Reference-based current chopping	Current-feedback adaptive PWM duty
Protection	Integrated protection circuits	UVLO and overcurrent merged into one FAULT path
Main contribution	Integrated motor driver and current sensing	Load-aware PWM that cuts duty as current rises, shown in simulation

The base paper's current regulation is not the same as our load-aware adaptive PWM; the extension is demonstrated in the simulation results that follow.

Limitations in existing work, and how we resolve them

Limitation in the literature	Our approach
Fixed, non-adaptive PWM or analog control	Adaptive PWM driven by sensed load current
Current sensing absent, or used only for protection	Integrated shunt sensing with amplifier and comparator
Adaptive techniques (dead-time) studied separately from load	Adaptation extended from dead-time toward load current
Protection limited or split across separate blocks	UVLO + OCP merged into one FAULT path (OTP: future work)
Little focus on robotic DC actuators	Targets brushed-DC robotic actuators with load-aware control

Research gap and proposed contribution

Existing literature

- Integrated H-bridge
- Integrated protection
- Current sensing (stepper)
- High-side gate driving
- Adaptive dead-time

Remaining gap

- Current sensing not combined with adaptive drive control
- Adaptive techniques studied separately
- Few designs target robotic DC actuators

Our direction

- Load-aware adaptive PWM
- Integrated current sensing
- Dead-time and protection in one hierarchy

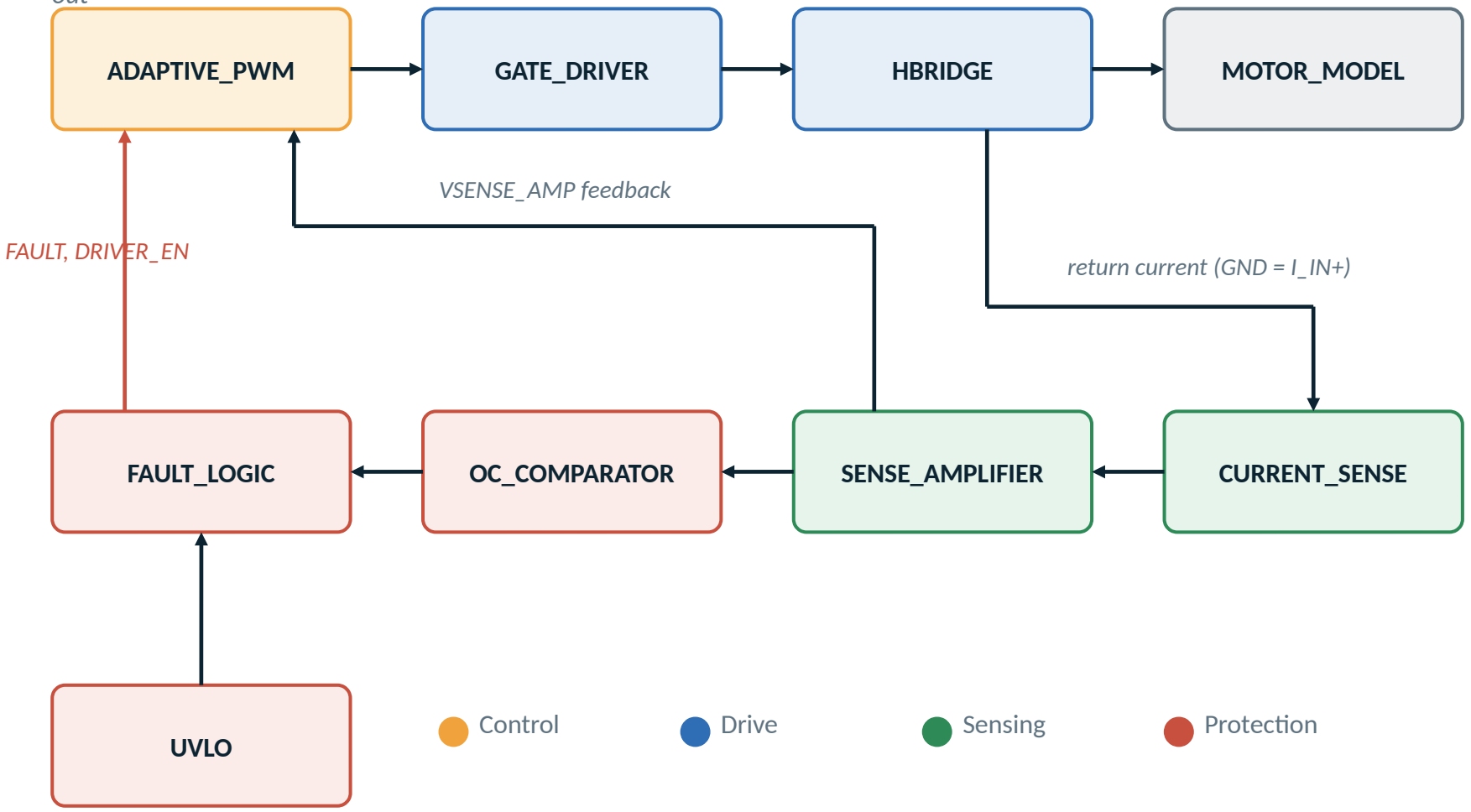
Proposed signal chain



Extension based on gaps identified in the five papers reviewed; novelty is claimed only against these.

Proposed architecture and hierarchy

Conceptual: DUTY_CMD, RAMP, DIR, DRIVER_EN, RESET in; FAULT_OUT, I_SENSE_OUT out



Hierarchy

- HBRIDGE
- MOTOR_MODEL
- GATE_DRIVER
- CURRENT_SENSE
- SENSE_AMPLIFIER
- OC_COMPARATOR
- UVLO
- FAULT_LOGIC
- ADAPTIVE_PWM
- ADAPTIVE_HBRIDGE_TOP

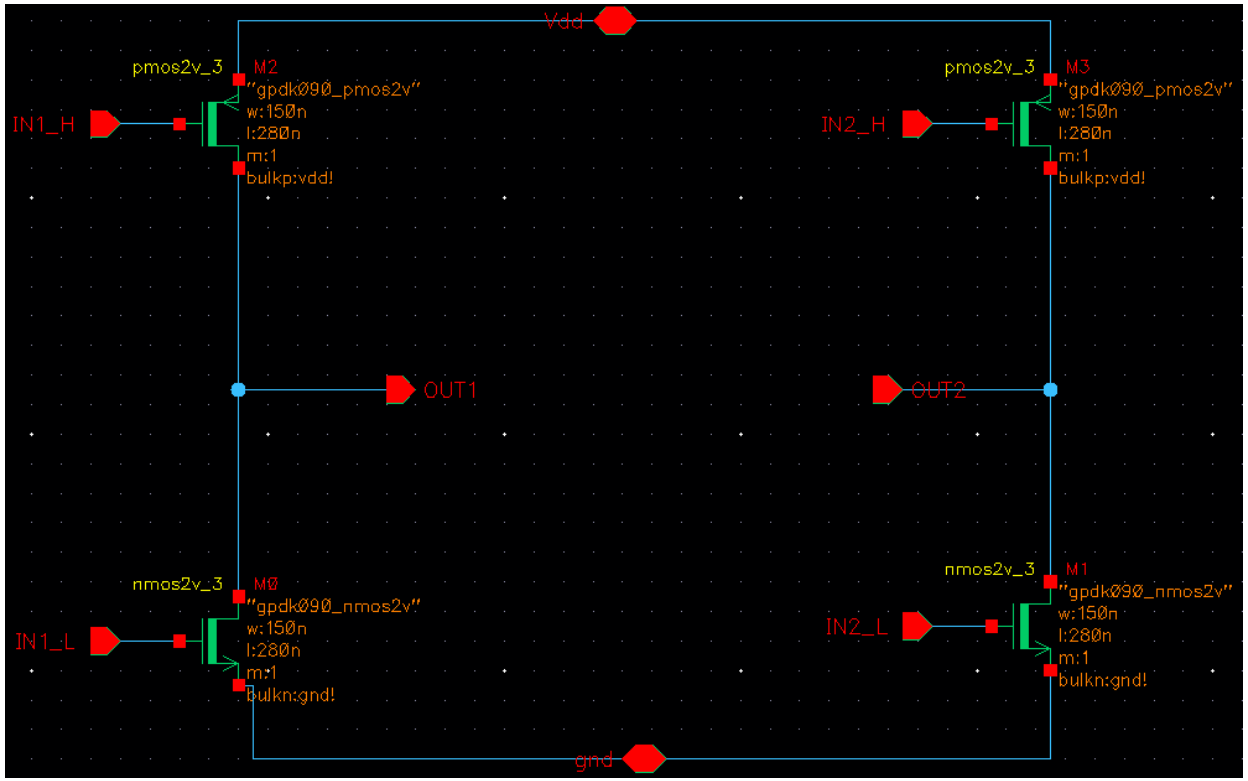
Actual Cadence hierarchy

- I0 Adaptive_PWM: OP_Amp, Comparator, CMOS_Inv, 2 NAND2, input RC networks
- I1 Protection: Bias_Reference, Overcurrent_Protection, Undervoltage_Lockout, Fault_Logic
- I2 Gate_Driver: 2 Dead_Time (one per leg), 2 Schmitt_Trigger, NAND2, CMOS_Inv, 4 Output_Buffer
- I3 Sense_Amplifier: OP_Amp with resistor and capacitor network
- I4 HBridge: 2 pmos2v_3 and 2 nmos2v_3 (gpdk090)
- I5 Current_Sense: shunt-based current sensing
- I6 Motor_Model: R-L motor load model
- I7 Bias_Reference
- Shared low-level cells: CMOS_Inv, NAND2, NOR2, Schmitt_Trigger, RC_Delay, Comparator, OP_Amp

What the hierarchy shows

- Eight instances sit under Adaptive_HBridge_Top; the other cells are lower-level building blocks
- Protection is a wrapper around Bias_Reference, Overcurrent_Protection, Undervoltage_Lockout and Fault_Logic
- Bias_Reference appears inside Protection and at top level
- Signal path as drawn: Adaptive_PWM, Gate_Driver, HBridge, Motor_Model
- Outputs at the top level: FAULT_OUT and I_SENSE_OUT
- Over-temperature protection is future work

H-bridge power stage

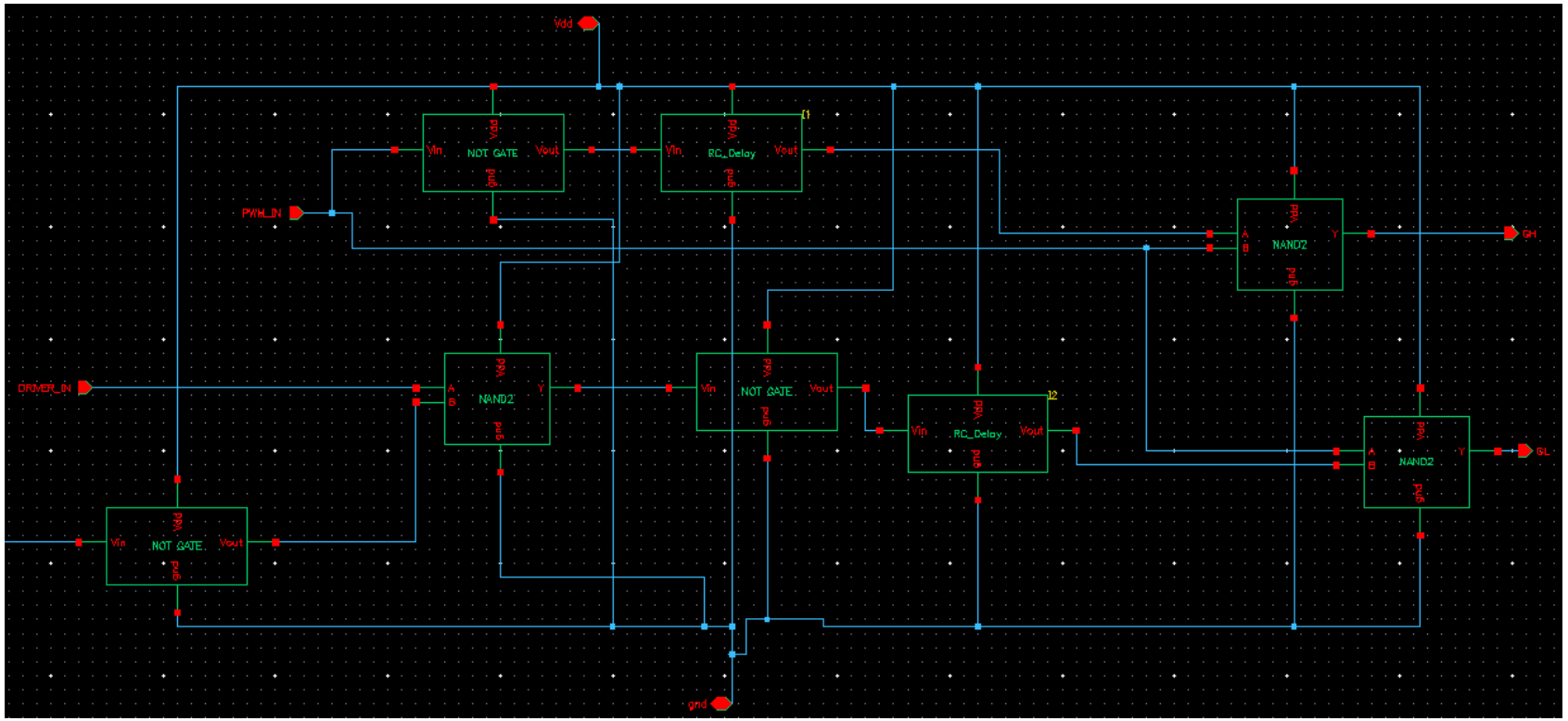


Switch states by mode (gate levels)

Mode	GH_A	GL_A	GH_B	GL_B
Forward	0	0	1	1
Reverse	1	1	0	0
Brake	1	1	1	1
Coast	1	0	1	0

- PMOS conducts when its gate is low; NMOS when high
- Leg 1: M2 (PMOS) over M0 (NMOS) to OUT1; leg 2: M3 over M1 to OUT2
- Never drive GH = 0 with GL = 1 on one leg (shoot-through)
- gpdk090 2 V devices, W = 150 nm, L = 280 nm; PMOS bulk to Vdd, NMOS bulk to gnd (confirmed from schematic)

Gate driver: enable, delay and gate logic



Enable
FAULT is inverted and combined with DRIVER_IN in a NAND2

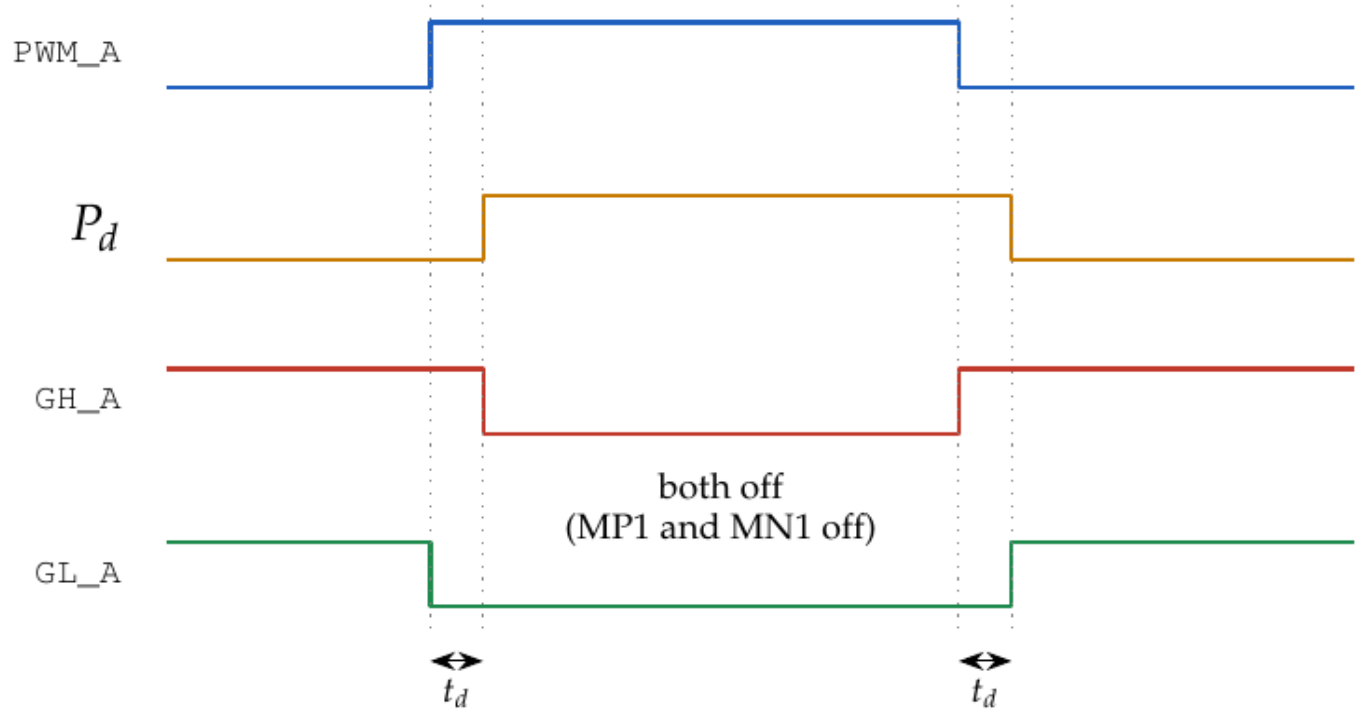
Dead-time delay
Two RC_Delay cells delay the PWM path to set the dead-time

Gate logic
Two NAND2 outputs form GH and GL; no level shifter is drawn

Dead_Time cell (Cadence), one per leg. Gate_Driver instantiates two of them plus four Output_Buffer cells.

Conceptual timing diagram of the dead-time scheme

Gate driver: dead-time operation



What the waveform shows

Rising edge: low side turns off at once; high side turns on after t_d

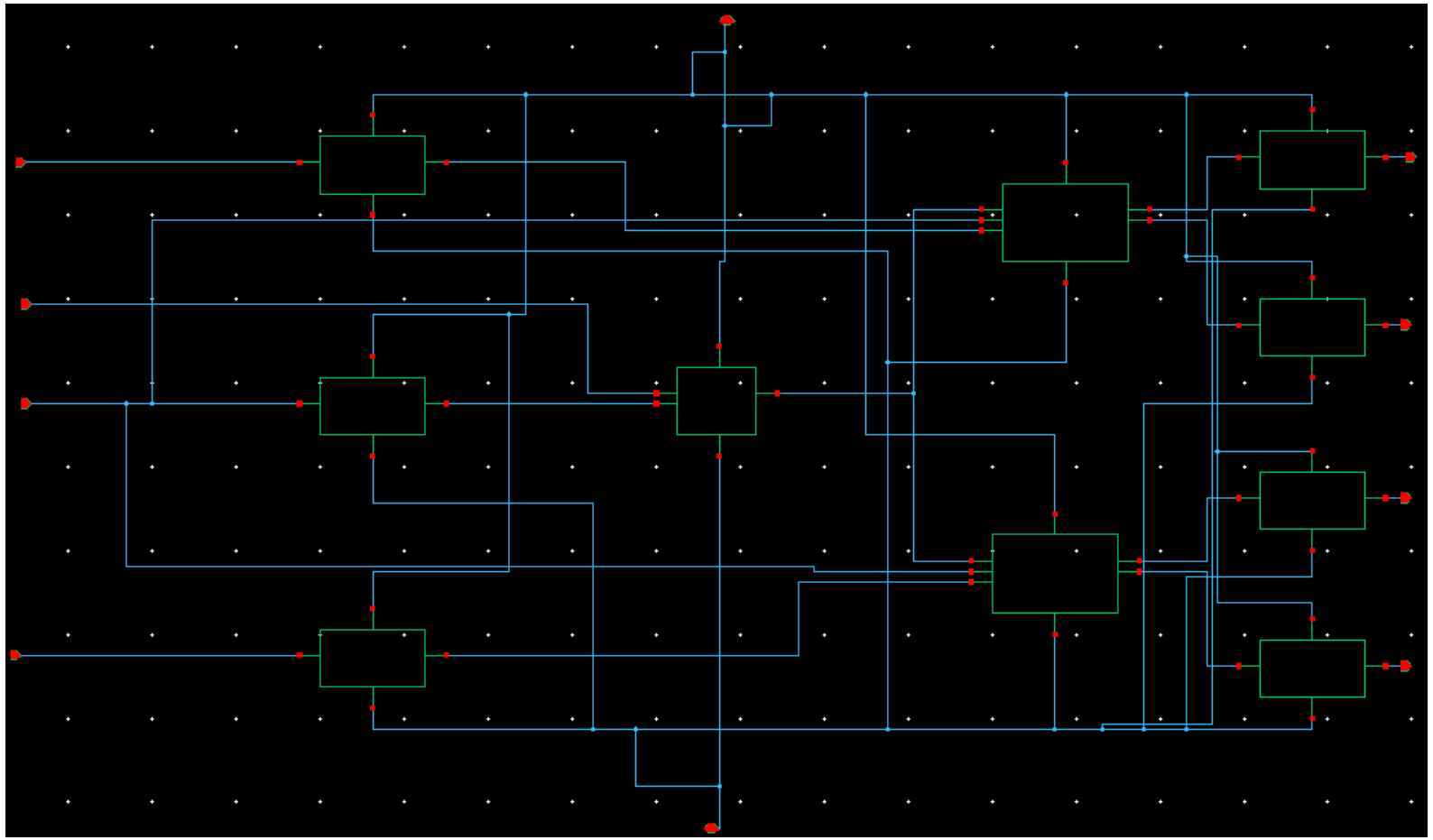
Falling edge: high side turns off at once; low side turns on after t_d

GH_A is active-low because MP1 is a PMOS

Design t_d about 100 ns = 0.69 Rd Cd (10 kOhm, 15 pF)

Fixed RC delay, too short to resolve on the 50 us plots; adaptive dead-time (paper 5) is future work

Gate_Driver hierarchy (Cadence)

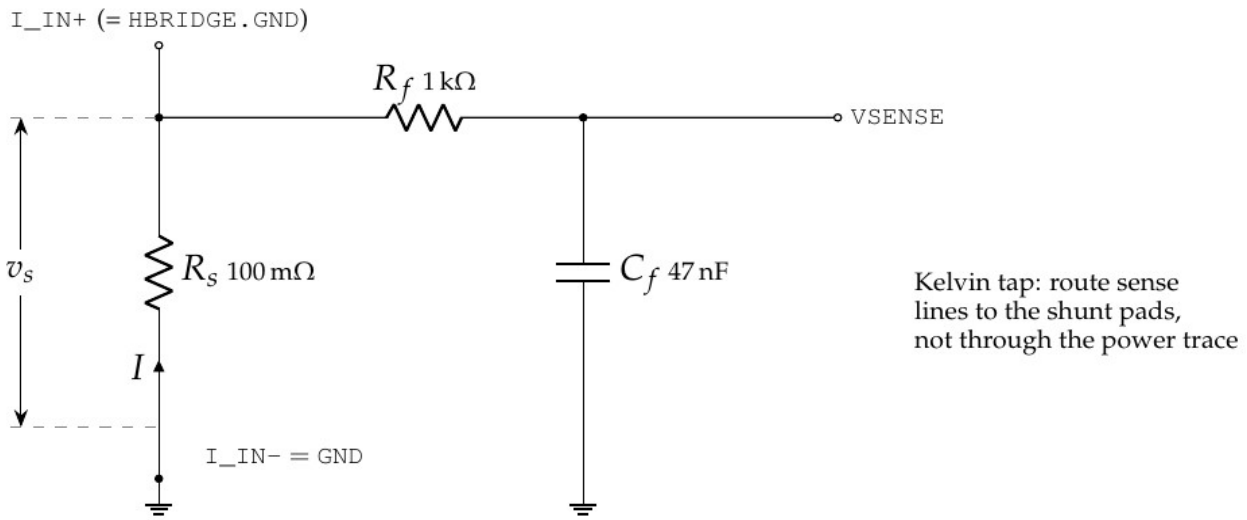


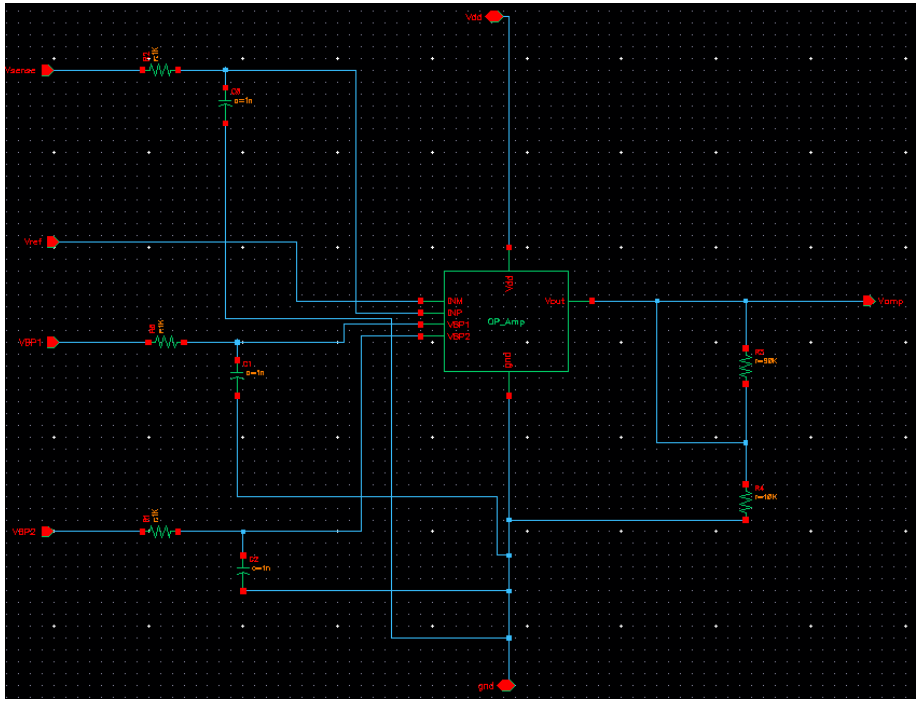
As drawn

- Two Dead_Time cells, one per leg
- Four Output_Buffer cells drive GH_A, GL_A, GH_B, GL_B
- Two Schmitt_Trigger cells, a CMOS_Inv and a NAND2 condition the inputs
- DRIVER_EN and FAULT enter the gate logic

Left: Current_Sense concept. Right: Cadence Sense_Amplifier

Current sensing and amplification

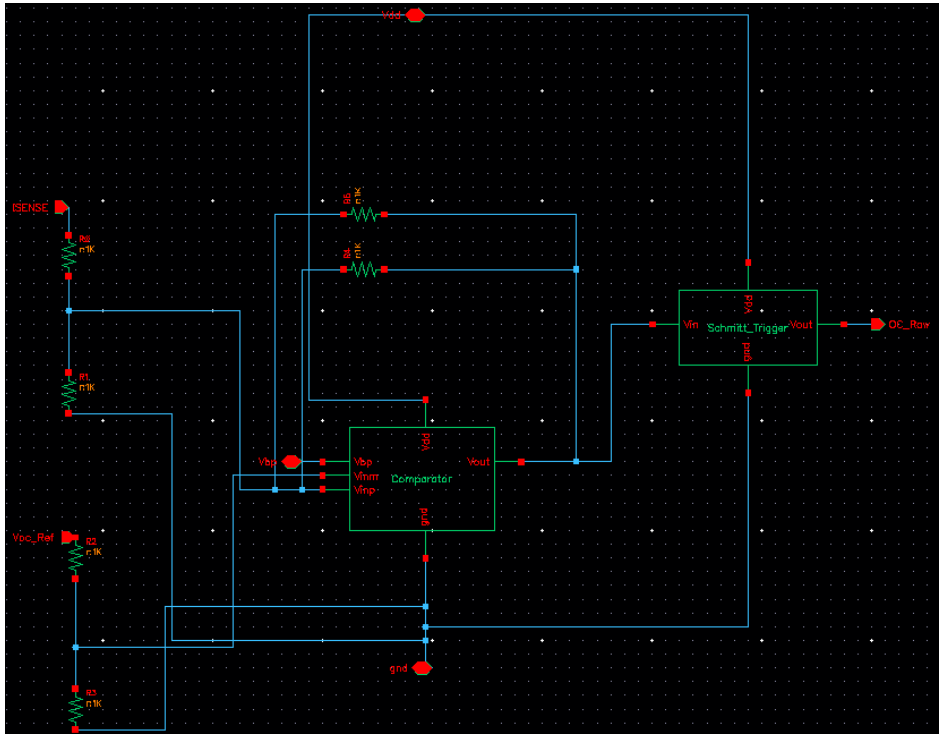


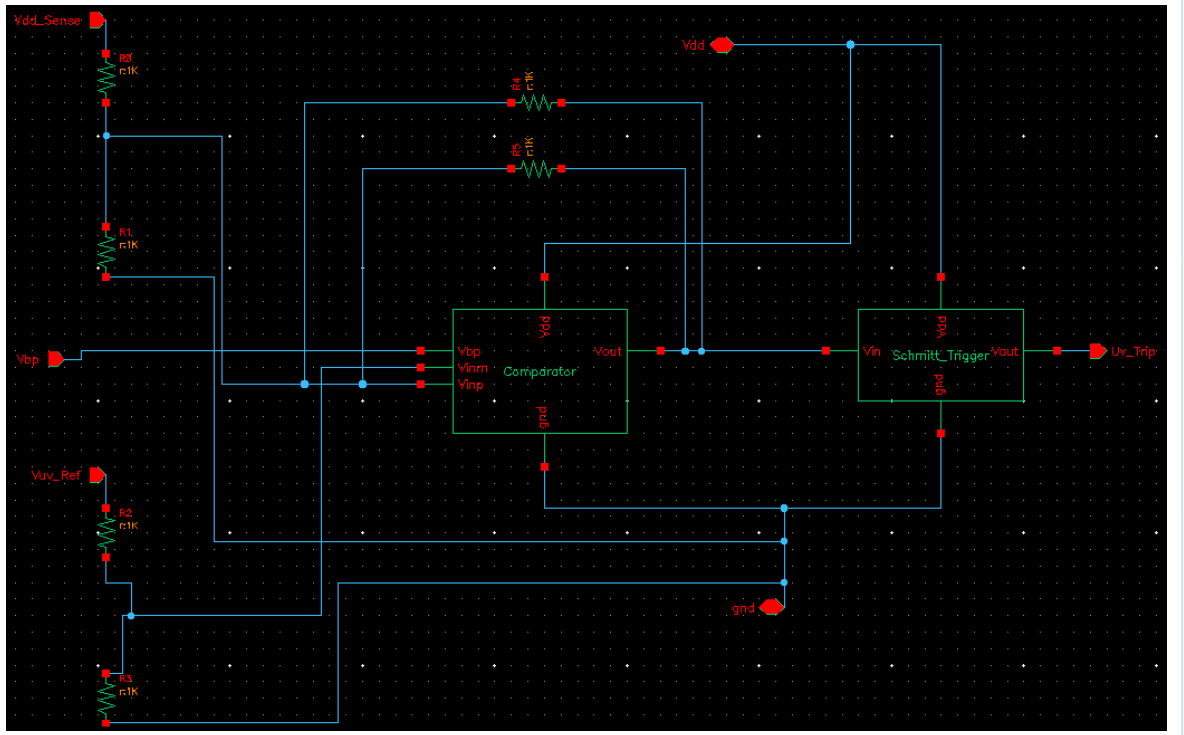


Current_Sense Design values: V_SENSE = I x Rs with Rs = 100 mOhm (0.1 V per amp). Shunt loss 0.27 W at 1.65 A. RC filter about 3.4 kHz.

Sense_Amplifier OP_Amp with RC-filtered inputs and a resistor feedback network that scales the shunt voltage into ISENSE_OUT and the feedback to Adaptive_PWM.

Overcurrent and undervoltage protection

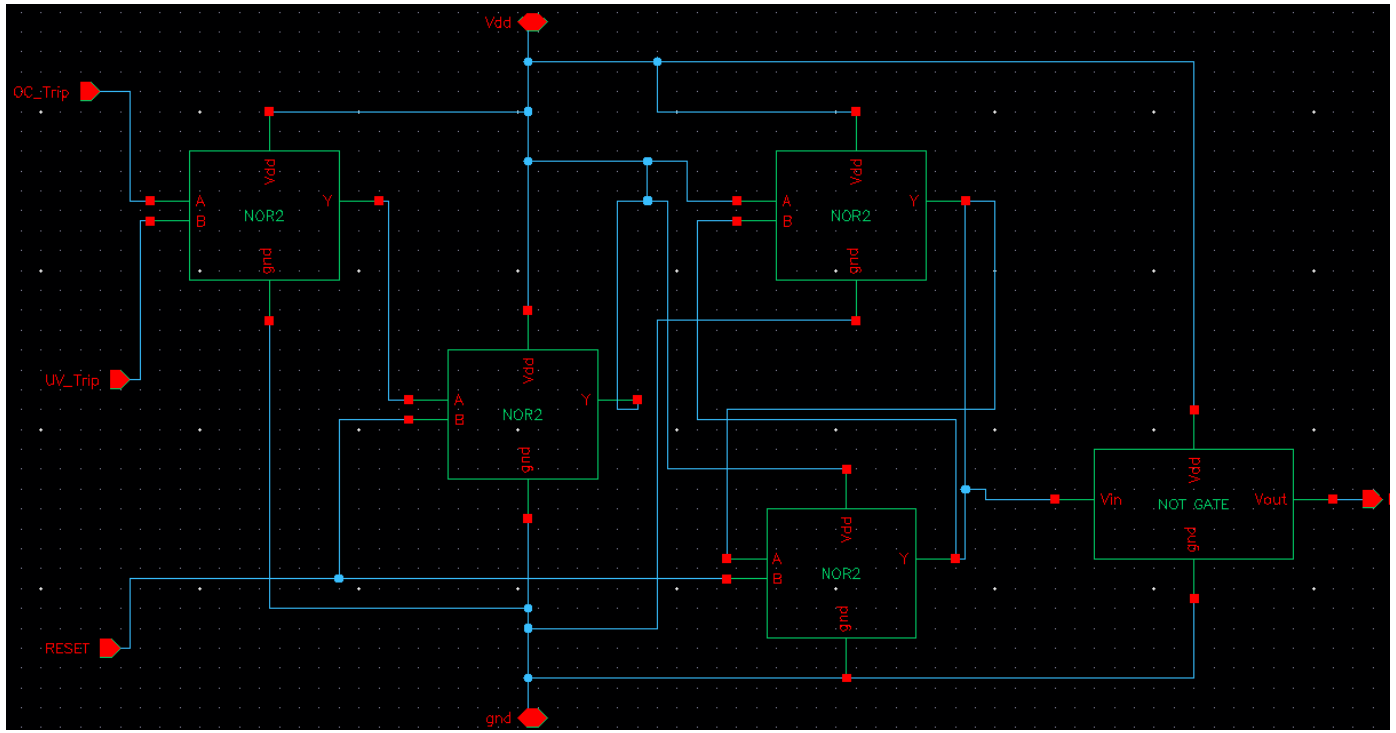




Overcurrent_Protection ISENSE and Voc_Ref dividers, Comparator with feedback resistors and Schmitt_Trigger give OC_Raw. The trip level is set by Voc_Ref (design target 1.65 A).

Undervoltage_Lockout Vdd_Sense and Vuv_Ref dividers, Comparator with feedback and Schmitt_Trigger give Uv_Trip. The thresholds are set by the dividers and Vuv_Ref.

Fault logic: latch with RESET



FAULT

SR-type latch of cross-coupled NOR2 gates: set by OC_Trip or UV_Trip, cleared by RESET

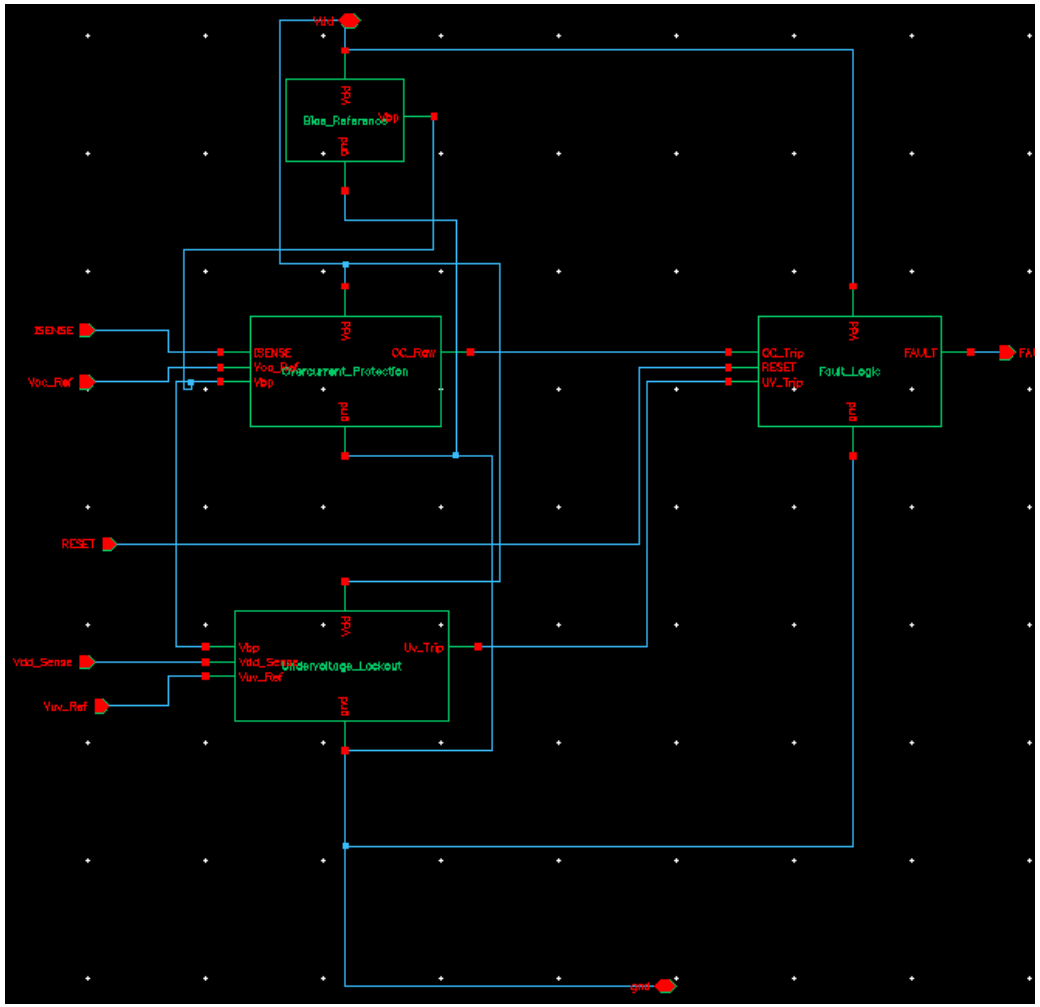
FAULT and DRIVER_EN

Both enter Gate_Driver; FAULT is gated with DRIVER_IN inside Dead_Time

Behavior

In the top-level run FAULT_OUT asserts once, about 58 to 62 us, then clears. No OTP input: OTP is future work.

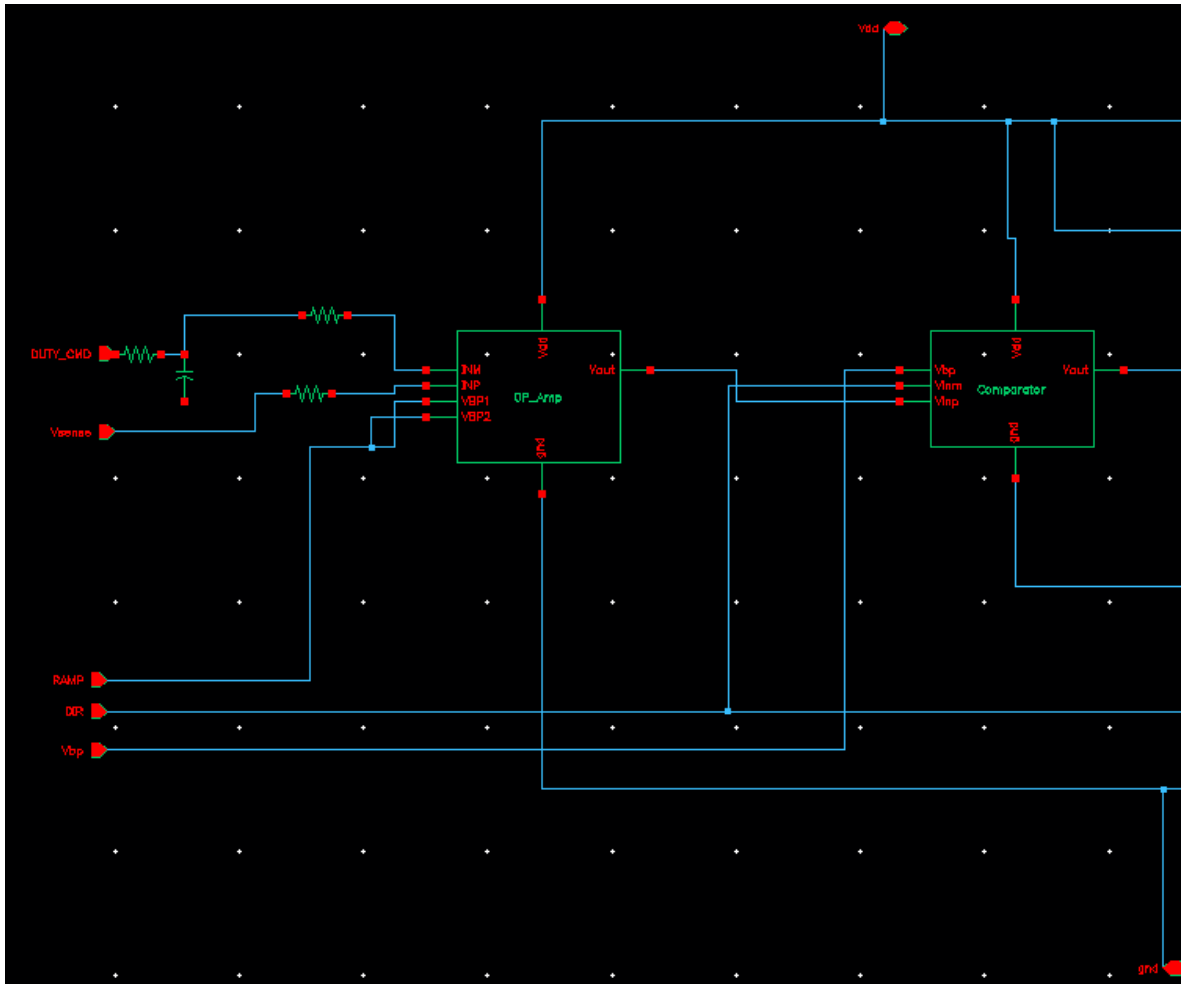
Protection wrapper (Cadence)



As drawn

- Wraps Bias_Reference, Overcurrent_Protection, Undervoltage_Lockout and Fault_Logic
- Inputs: ISENSE, Voc_Ref, Vdd_Sense, Vuv_Ref, RESET
- OC_Raw and Uv_Trip feed Fault_Logic; FAULT is the only output
- Voc_Ref and Vuv_Ref are the reference inputs that set the trip points

Adaptive_PWM: op-amp stage



What the schematic contains

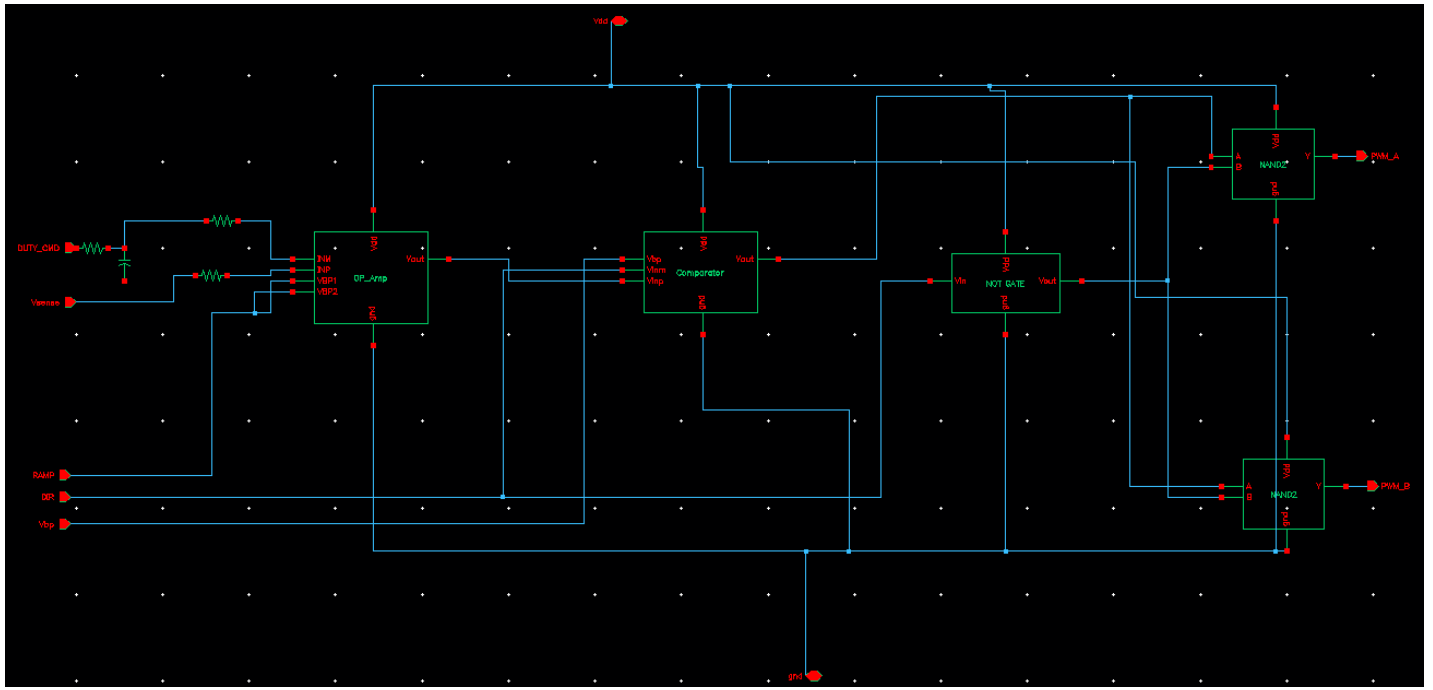
OP_Amp takes DUTY_CMD (through an RC network) and Vsense (through a resistor); VBP1 and VBP2 are bias pins

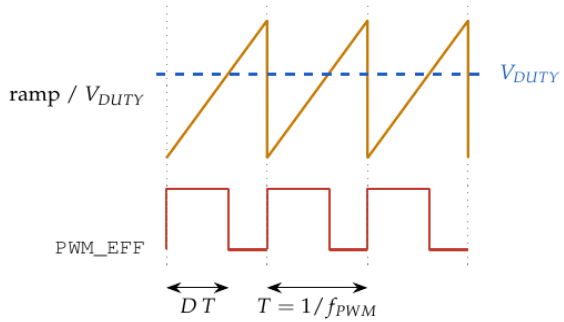
Its output feeds the Comparator on the next slide

The effect on PWM duty is shown in the adaptive-response results

Right: conceptual PWM sketch

Adaptive PWM: modulator and direction steering



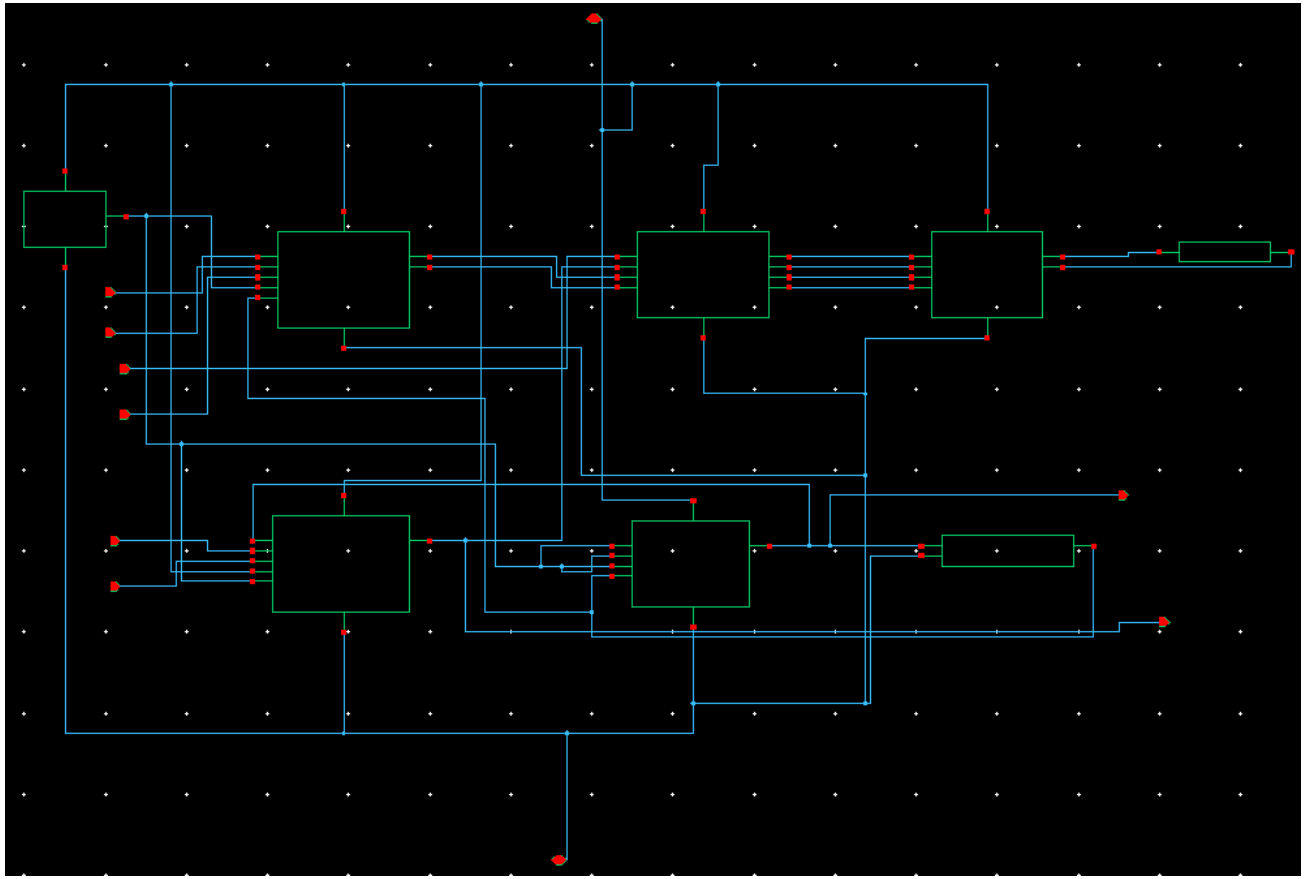


Modulator
Comparator output sets the effective PWM; ramp in the top-level testbench is 0 to 1 V, 10 us period

Direction
DIR feeds a CMOS_Inv and two NAND2 gates forming PWM_A and PWM_B

Fault disable
Adaptive_PWM has no FAULT or enable port; disabling happens in Gate_Driver

Top-level integration

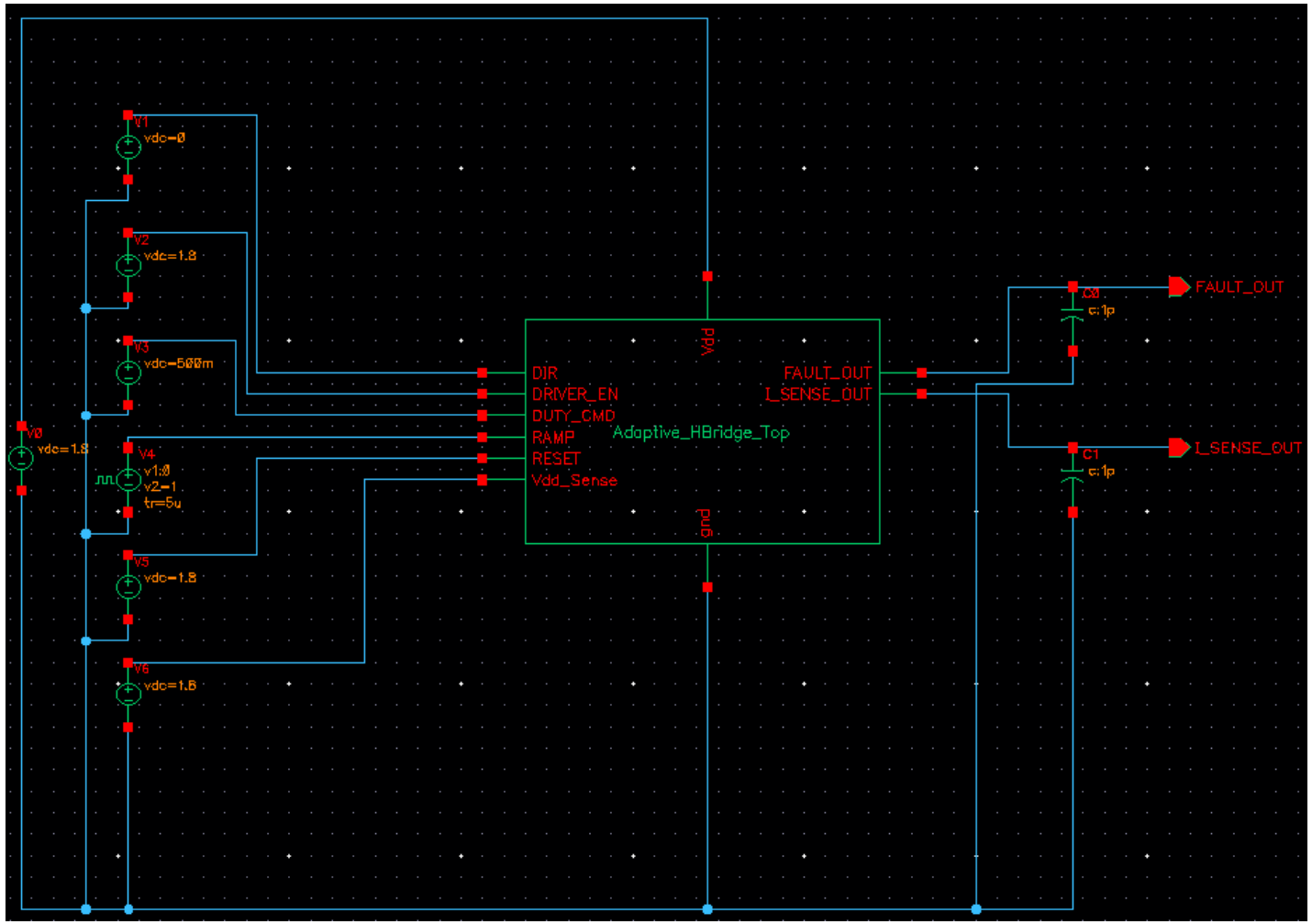


Top-level ports

Port	Role
DIR	input: direction select
DRIVER_EN	input: driver enable
DUTY_CMD	input: analog duty command
RAMP	input: modulator ramp
RESET	input: fault reset
Vdd_Sense	input: supply sense for UVLO

Outputs: FAULT_OUT (fault flag) and I_SENSE_OUT (sensed current).

Testbench: TB_Adaptive_HBridge_Top



Sources as drawn

- V0: supply, 1.8 V
- V1: DIR, 0 V
- V2: DRIVER_EN, 1.8 V
- V3: DUTY_CMD, DC, about 0.5 V
- V4: RAMP, pulse 0 to 1 V
- V5: RESET, 1.8 V
- V6: Vdd_Sense, 1.8 V
- 1 pF loads on FAULT_OUT and I_SENSE_OUT

Design parameters

Parameter	Value	Basis
Logic supply VDD	1.8 V (V0, HBridge Vdd)	Schematic
Power devices	gpdk090 2 V, W = 150 nm, L = 280 nm	Schematic
PWM ramp	0 to 1 V, 10 us period (top-level run)	Waveform
Duty command	DC, about 0.5 V	Testbench
Simulated current	about 0.6 A to 1.7 A	Waveform
Shunt Rs	100 mOhm	Design value
Dead-time	about 100 ns (0.69 Rd Cd)	Design value
Overcurrent reference	1.65 V (1.65 A)	Design target
Current-limit knee	1.2 A	Design target
PWM duty range	0-90 %	Design target

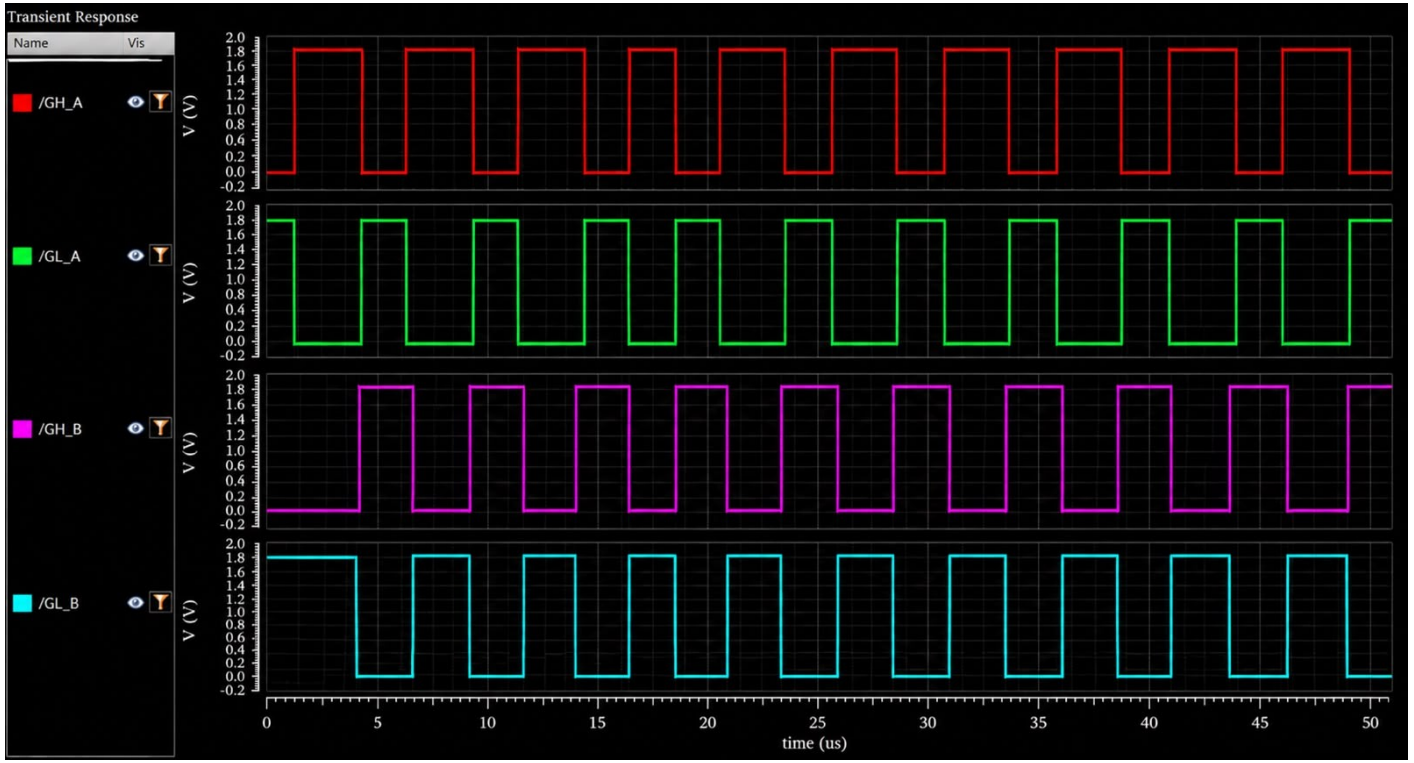
Device and supply limits

The design is simulated with gpdk090 2 V devices at a 1.8 V supply, with motor current up to about 1.7 A. The 1-5 A and 6-24 V range needs high-voltage devices and is future work.

Protection thresholds

UVLO thresholds (8.0 V fall / 8.5 V rise) belong to the 6-24 V target. The 1.8 V testbench does not exercise them.

Results: gate driver outputs



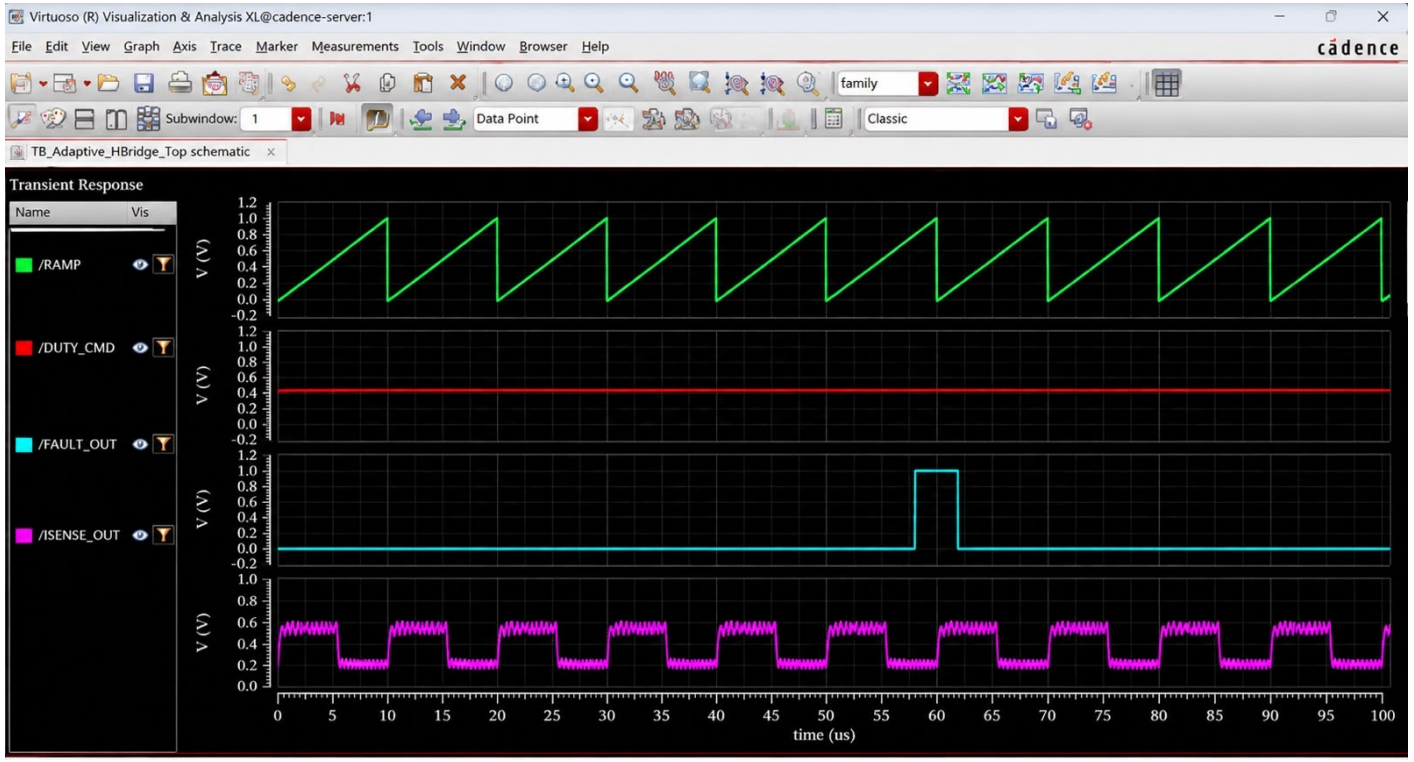
What the waveforms show

- GH_A, GL_A, GH_B and GL_B all swing the full 0 to 1.8 V logic supply
- On each leg GL is the complement of GH, so the two devices of a leg are driven in opposition
- Leg B follows leg A by about 3 us (GH_B first rises near 4.2 us, GH_A near 1.3 us)
- Switching period is about 5 us in this 50 us run

The 100 ns design dead-time is below the resolution of this plot.

Result The Gate_Driver turns the PWM inputs into full-swing GH / GL drive for both H-bridge legs.

Results: top-level testbench



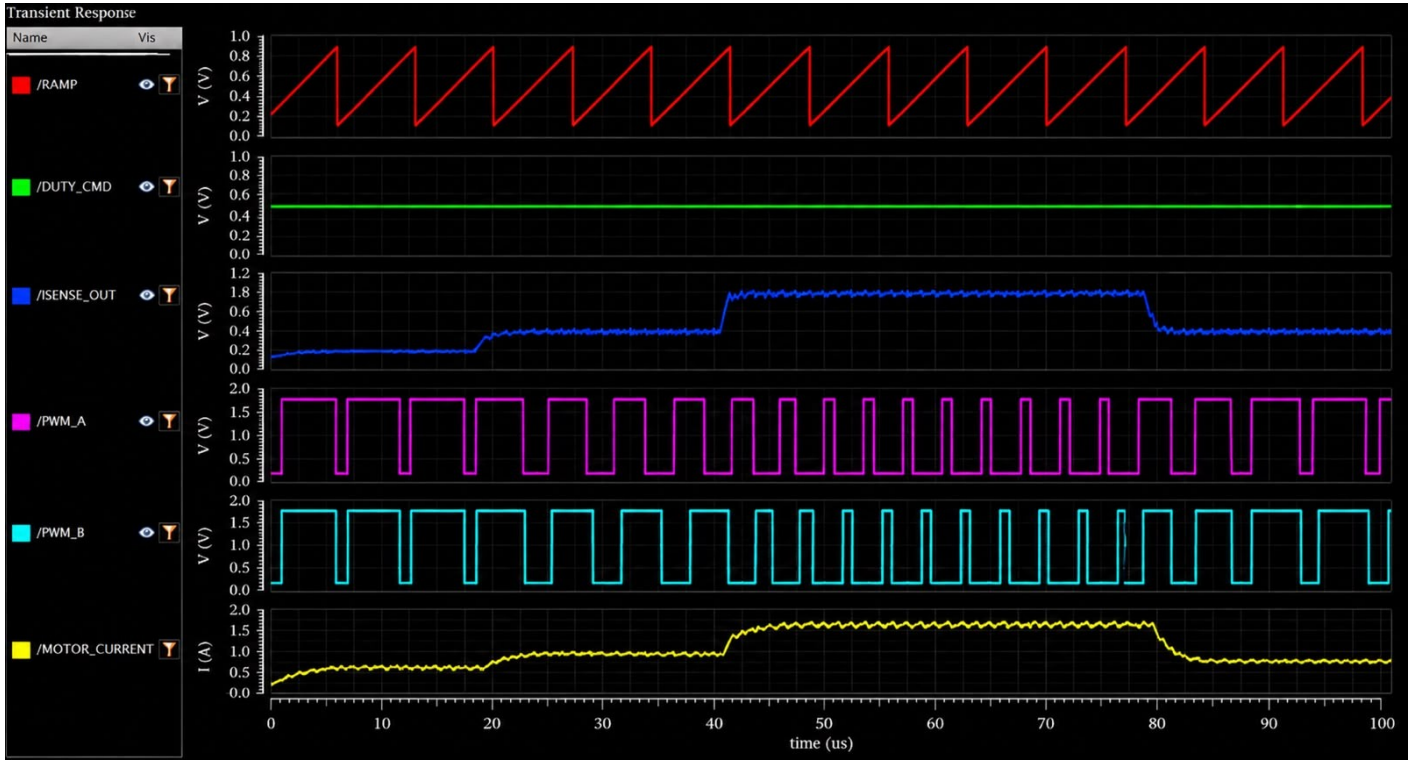
What the waveforms show

- RAMP: 0 to 1 V sawtooth, 10 us period
- DUTY_CMD: constant, about 0.45 V
- ISENSE_OUT: alternates between about 0.6 V and 0.2 V once per PWM cycle, with ripple on both levels
- FAULT_OUT: one pulse to about 1 V from about 58 to 62 us, then clears

Sensed current follows the PWM cycle; the fault flag asserts once and releases.

Result Ramp, duty command, sensing and fault paths run together in the top-level testbench.

Results: PWM duty versus load current



What the waveforms show

- DUTY_CMD stays fixed at about 0.5 V for the whole 100 us run
- MOTOR_CURRENT steps from about 0.6 A to 0.95 A (19 us), 1.7 A (41 us), then back to 0.8 A (80 us)
- ISENSE_OUT follows each step: about 0.18 V, 0.4 V, 1.8 V, then 0.4 V
- PWM_A and PWM_B high-time falls from roughly 85 % to 35 % of the cycle between 41 and 79 us, then widens again

At a fixed duty command, PWM duty drops as sensed current rises.

Duty percentages are read from the plot.

Result The loop cuts PWM duty when sensed load current rises and restores it when the load falls: the load-aware behaviour this project adds.

Comparison with previous work

Feature	1 Zhang	2 Xu	3 Krenzke	4 Dong	5 He	Ours
Technology	0.5 um HV CMOS	0.18 um CMOS	0.35 um CMOS	0.18 um BCD	0.18 um BCD	**GPDK 90 nm (gpdk090), 2 V devices**
Motor	DC	BLDC vibration	DC (general)	2-phase stepper	DC / BLDC	**DC robotic actuator**
Supply	2.5-5.5 V	2-4 V	up to 36 V	8-40 V	5.2 V / 24 V in	**1.8 V (simulated)**
Max current	1 A	300 mA	external FETs	2 A (1.6 A cont.)	gate driver	**about 1.7 A (simulated)**
Control	PWM + analog	PWM (Hall)	PWM to 50 kHz	current chopping	adaptive dead-time	**load-adaptive PWM (current feedback)**
Current sensing	basic	none	protection only	cascode mirrors	none	**shunt + amplifier, used for feedback**
Dead-time	-	-	fixed	-	adaptive	**fixed RC (adaptive: future work)**
Protection	UVLO, OCP, OTP	UVLO, thermal	cross-conduction, short	OCP, UVLO, OTP	cross-interlock	**UVLO, OCP, FAULT path (OTP: future work)**
Robotic DC target	No	No	No	No	No	**Yes**

Literature columns as summarized from the papers; the Ours column reflects the simulated design at 1.8 V (gpdk090).

Conclusions and future work

Conclusions

- Sensing, control, gate drive and protection are integrated in one Cadence hierarchy of eight instances
- Built on Dong et al. 2025: its integrated current sensing is extended to a brushed-DC actuator with load-aware PWM
- At a fixed duty command, PWM duty falls from roughly 85 % to 35 % as motor current rises from about 0.95 A to 1.7 A
- None of the five reviewed papers combines current sensing with load-adaptive drive

Future work

1. Scale to 1-5 A and 6-24 V (HV devices)
2. Adaptive dead-time (paper 5 technique)
3. On-chip current sensing (cascode mirrors)
4. Over-temperature protection
5. Real motor and robotic load validation

References

1. Zhang et al., "Design of Fully-Integrated H-Bridge for DC Motor Driver IC, 0.5-um HV CMOS," 2010.

2. Xu et al., "Monolithic H-bridge brushless DC vibration motor driver with a highly sensitive Hall sensor," 2012.

3. Krenzke et al., "A 36-V H-bridge driver interface in a standard 0.35-um CMOS process," 2003.

4. D. Dong, J. Leng, Z. Li, B. Lei, "A fully integrated stepper motor driver with a novel integrated current sensing technology," IEICE Electronics Express, vol. 22, no. 10, art. 20250071, 2025, doi:10.1587/elex.22.20250071.

5. J. He, H. Yu, D. Dong, D. Xie, M. Huang, adaptive dead-time high-side gate driver, Microelectronics Journal, vol. 167, 2026, Art. 106937, doi:10.1016/j.mejo.2025.106937.